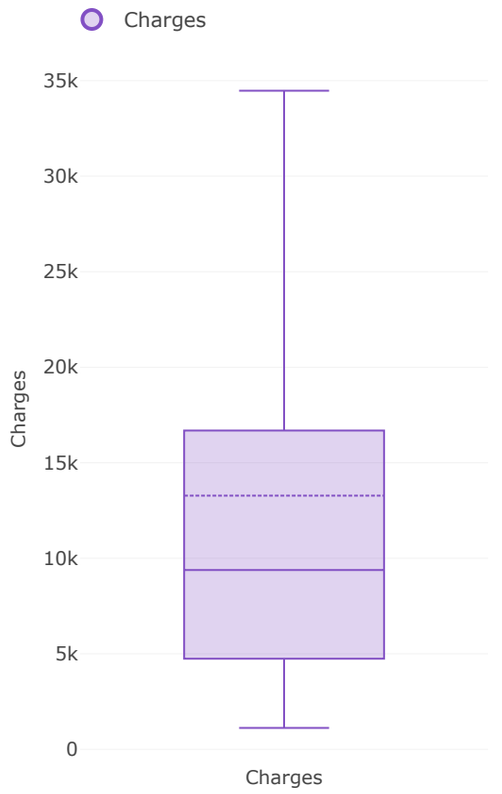
What factors have the strongest influence on medical insurance charges, and how do charges differ between smokers and non-smokers across age groups and regions?

1338
Number of Beneficiaries

$13,270.42
Average Charges

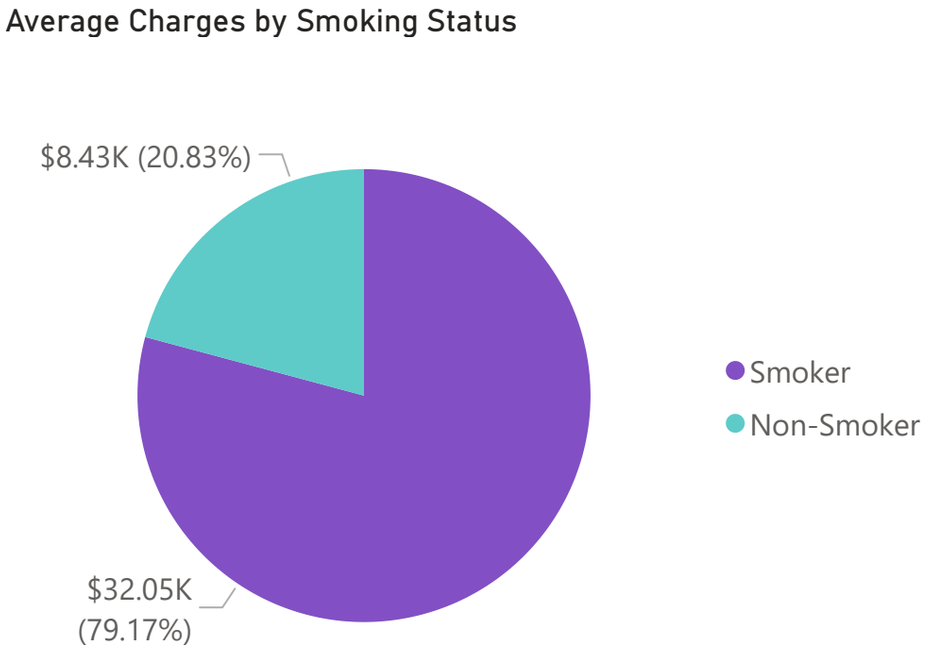
Region	Number
southeast	364
northwest	325
southwest	325
northeast	324
Total	**1338**

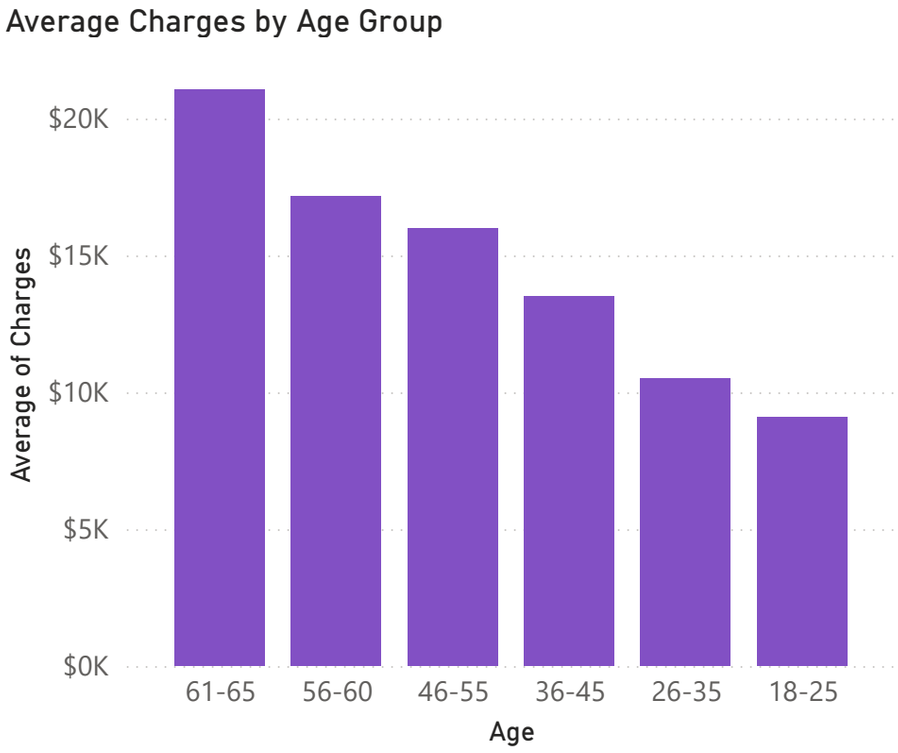
Descriptive Statistics of Charges



Created by: Alrayyan-A on GitHub https://github.com/Alrayyan-A

The Strongest Factors are Smoking and Age:

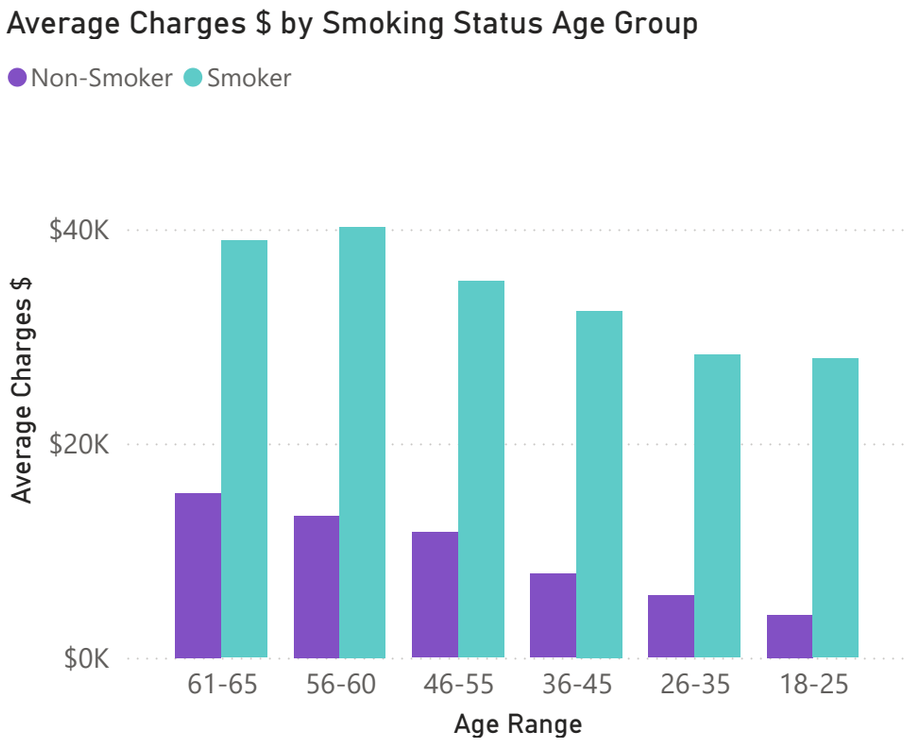




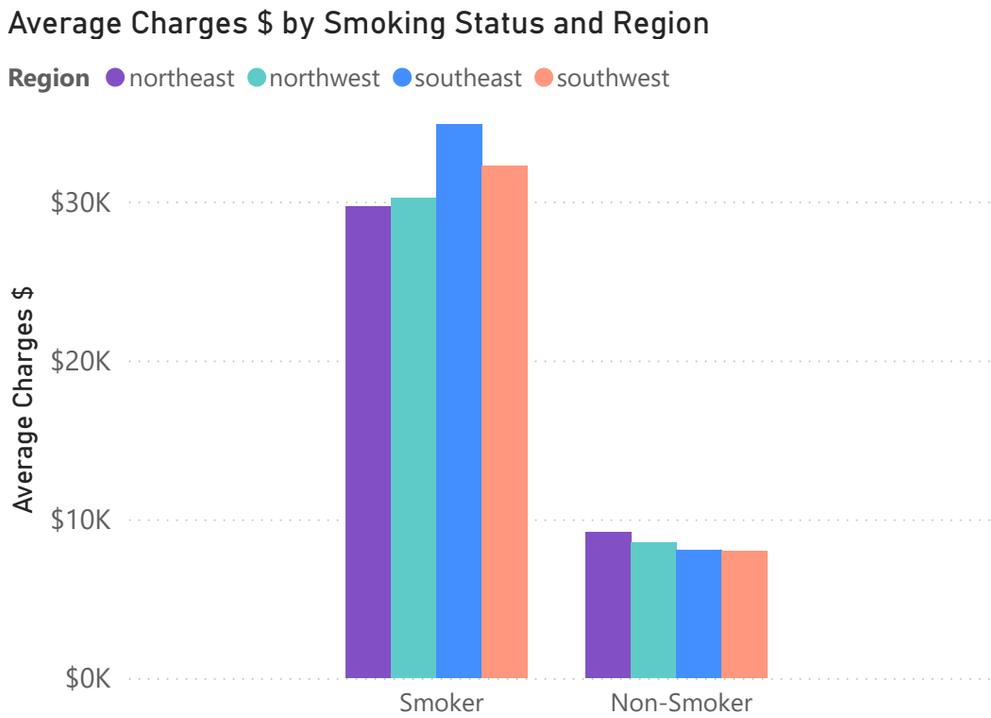
Smokers have **280%** higher medical insurance charges compared to non-smokers.

People aged 61-65 incur **65.83%** higher average insurance charges compared to the weighted average of charges of the other age groups.

Created by: Alrayyan-A on GitHub https://github.com/Alrayyan-A

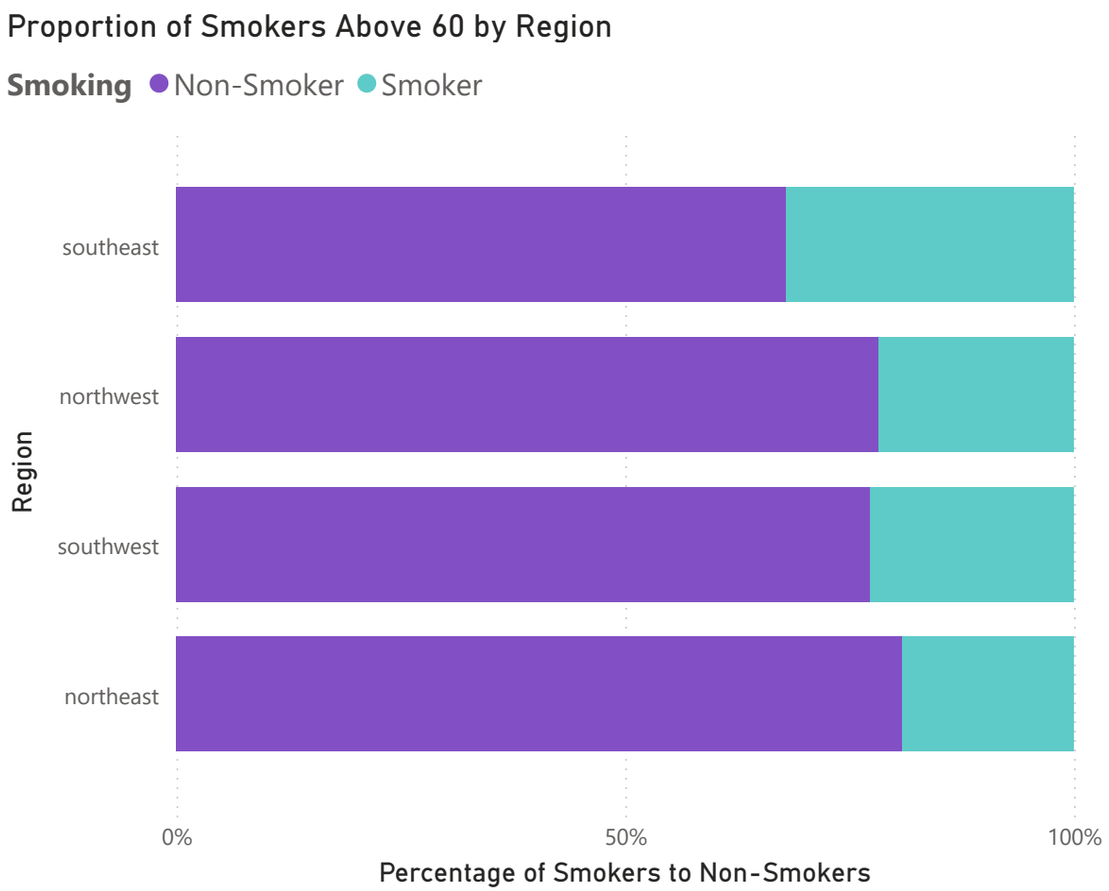


Among smokers, individuals aged 56-60 incur the highest charges, while individuals aged 61-65 incur the highest charges among non-smokers.



Among smokers, average charges are highest in the Southeast (**13.65%** above other regions) and lowest in the Northeast.

Non-smokers face their highest charges in the Northeast and their lowest in the Southwest, with less pronounced regional variation.



The southeast region has the highest proportion of smokers to non-smokers above 60 years old at **32%**, while the northeast is the lowest at 19.05%.

Created by: Alrayyan-A on GitHub https://github.com/Alrayyan-A

Conclusion:
Smoking and old age are strongly associated with high medial insurance charges. The southeast has the highest proportion of smokers to non-smokers.

Recommendations:
Market to young non-smokers, ideally, aged 18-35 from the northeast as they have smallest risk. In addition, organize preventative health programs for smokers.